



Stanford University

JAZLYN MOONEY

Postdoctoral Fellow

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December 15, 2020

Department of Quantitative and Computational Biology
University of Southern California
1050 Child Way, RRI 201 (MCB)
Los Angeles, CA 90089-2910

Dear Prof. Rohs and Members of the Search Committee,

I am applying for the tenure-track faculty position in Computational Biology at the Assistant Professor rank in the Department of Quantitative and Computational Biology at the University of Southern California. I completed my PhD in Human Genetics at the University of California, Los Angeles, in September 2020, and since then have held a postdoctoral position at Stanford. My research combines computational approaches with population genetics theory and evolutionary anthropology to better understand genetic variation, medical genetics, and human evolution.

Human health is determined through the interaction of many factors, including social, demographic, and evolutionary processes. My future research program will be focused on using methods development, theoretical population genetics, and simulations to understand human evolution and how a population's history influences the prevalence of disease-associated variants. Further, I will also focus on developing methods to improve detection of deleterious disease-associated variants. Most importantly, I will continue research on understudied populations in genomics. As the representation of diverse populations is imperative for ameliorating the diversity imbalances in current genomics data sets and preventing further exacerbation of disparities that already exist within medicine and genomics.

My previous research has integrated large-scale whole genome sequence datasets and genotype data from both human and *Canid* populations, species with distinct demographic histories, to develop a more complete picture of mechanisms that shape patterns of deleterious variation. I examined the joint impact of demography and natural selection on genetic variation in order to determine how these forces have allowed for deleterious variation to accumulate in populations. My work in human populations was focused on how demography affects the distribution of deleterious variation in a young admixed population isolate. While most studies of population isolates have focused on populations where the founders all came from the same ancestral population, we investigated genomic diversity of recently admixed population isolates from Costa Rica and Colombia and compared their diversity to a benchmark population isolate, the Finnish. I also used *Canids* (breed dogs, village dogs, and wolves) as a model system to test how recent demography and artificial selection affect the distribution of deleterious variation and the architecture of complex traits and identify regions of the genome that are potential hotspots for inbreeding depression.

My work has collectively yielded 2 peer-reviewed research articles (as well as an article in review at PNAS), including a first-authored publication in the American Journal of Human Genetics, the flagship journal of the American Society of Human Genetics. My work has also been selected for talks at internal conferences such as the Informatics Center for Neurogenetics and Neurogenomics Symposium and larger conferences such as The Society for Molecular Biology and Evolution (SMBE) Meeting. In addition to research experience, I also am

practiced in securing external funding, including the National Science Foundation Graduate Research Fellowship.

Alongside of my academic research, I will continue to spend time cultivating relationships with other underrepresented faculty and students in STEM through mentorship, teaching, and actively participating in diversity efforts at USC. I have experience with working across departmental home areas and coordinating efforts with students and their departmental directors to build infrastructure for funding, create organizations, and to implement student initiatives into umbrella programs. I have also led and coordinated workshops on basic data analysis and genomic data analysis within the department and at conferences such as Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS).

When teaching and mentoring, I believe that a student's ability to connect with an individual as a scientific role model is a key-component for the persistence and retention of diverse scientists in STEM. When teaching, I can envision contributing to existing courses such as Practical Analysis of Biological Data in R, Computational Genome Analysis, or Introduction to Genome Science, as well as creating new courses such as a research colloquium or a course on the history of genomics. The research colloquium would foster both cross-departmental collaboration and introduce students to faculty outside of USC. Speakers would not only cover their research, but also cover their career paths allowing students to meet potential scientific role models. A course on the history of genomics, would encompass the many discoveries in the field of genetics, which have had great impact on understanding the natural world and human diseases. We would also discuss the eugenic origins of genetics, which has unfortunately led the field to be responsible for great evil. This course can be tailored for both undergraduate and graduate students. The goal being to discuss these topics through selected academic readings and research, as well as where we as leading discussions on where scientists can go next to improve equity in genomics.

The interdisciplinary focus of this department makes this position particularly appealing to me. My research on human evolution and population genetics theory would complement the departments strong background in computational biology and bioinformatics. I look forward to the possibility of collaborating with faculty in the Dornsife College of Letters, Arts and Sciences, the Viterbi School of Engineering, and in particular the Keck School of Medicine. Please find attached to this letter my complete application materials. I have contacted Noah Rosenberg, Nelson Freimer, Kirk Lohmueller, and Gregory Payne (to speak about my diversity, equity, and inclusion efforts) to submit reference letters on my behalf upon request. Thank you for your time. I look forward to hearing from you.

Sincerely,

A handwritten signature in black ink that reads "X Jazlyn A. Mooney". The signature is written in a cursive, flowing style. The "X" is at the beginning, followed by "Jazlyn A." and then "Mooney". A horizontal line is drawn across the middle of the signature, passing under the "Jazlyn A." part.

Jazlyn Mooney

JAZLYN MOONEY
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EDUCATION

Stanford

Postdoctoral Fellow

Advisor: Noah Rosenberg

2020-Present

University of California Los Angeles

Ph.D. in Genetics and Genomics

Advisor: Kirk Lohmueller

2015-2020

University of New Mexico

BS with Honors Anthropology and Biology (Summa Cum Laude)

Honors Thesis: "Inferring Evolutionary History from Ancestry Informative Markers (AIMs)"

Advisor: Jeffrey Long

2010-2014

RESEARCH INTERESTS

I am interested in using bioinformatics to analyze next-generation sequence data to study patterns of human genetic diversity across the world. Additionally, I am interested in exploring how we can implement population genetics to better understand the evolutionary origins of disease in human populations.

RESEARCH EXPERIENCE

DNA Science Intern at Ancestry.com

Mentor: David Turissini

DNA Science summer intern with the Population Genetics team at Ancestry.com in San Francisco.

Project title: Generating Hierarchical Population Labels Informed by Genetics

June 2019-August 2019

University of California Los Angeles Genetics and Genomics Department

Advisor: Kirk Lohmueller

2015-2020

1. My first project focused on determining the impact of recent demography on genetic diversity in admixed population isolates. I analyzed whole genome sequence data from Costa Rican and Colombian isolates and compared them to the Finnish population. Additionally, I determined the amount of genetic diversity in each group by using population-genetic summary statistics, tested mechanisms that generate long runs of homozygosity, and analyzed the distribution of deleterious variation in the genome. Finally, I explored the relationship between pedigree inbreeding coefficients, ancestry, long runs of homozygosity, and fitness.
2. My second project is focused on conservation genomics and testing the impact of long-term small effective population size in Ethiopian Wolves. I inferred the demographic history of Ethiopian Wolves using approximate Bayesian computation. I surveyed genetic diversity in the Ethiopian Wolf relative to other wolves and breed dogs. Lastly, I have run a selection scan to determine whether there are genes under positive selection to aid adaptation to high altitude.
3. My third project involved using dogs as a model organism for examining the link between complex disease and weakly deleterious recessive variants. I did this by associating the burden of long runs of homozygosity and identity-by-descent segments with a number of different phenotypes. I also examined biases in the Online Mendelian Inheritance in Animals database, which contains a list of Mendelian disease causal variants and the breed in which they were discovered. Finally, I tested whether long runs of homozygosity can be used to detect inbreeding depression in a population.

University of New Mexico Evolutionary Anthropology Department**2012-2014****Undergraduate Research Assistant****Advisor: Jeffrey Long**

My research was focused on both data processing and analysis. My data processing projects involved developing and writing a protocol for downloading and working with sequence data from the UCSC Genome Browser. I also made a pipeline to assemble genomes and produce comparative phylogenetic trees as the final output.

For the data analytics portion of my research, I filtered HapMap genome data for target African and Non-African ancestry informative markers, AIMs. I then constructed haplotypes containing the target AIMs and their flanking SNPs. I calculated the homozygosity for each haplotype and used this information to identify potential archaic introgression in Non-African populations.

University of Chicago - Pritzker School of Medicine Experience in Research (PSOMER)**2013****Undergraduate Research Assistant****Advisor: Michael Maitland**

My research project focused on using genetic haplotypes to explore patient reactivity to Sunitinib and Sorafenib, drugs used to treat advanced cancer. Specifically, I looked in African American males for SNPs that were correlated with improved treatment outcome. I presented my research and poster at an internal conference at the end of the eight weeks.

University of New Mexico - Research in Molecular Biology**2013****Undergraduate Research Assistant****Advisors: Donald Natvig and Coenraad Adema**

This research integrated both benchwork and computational work. First, I processed DNA and RNA sequences from both plants and animals using basic purification techniques that are required for collecting sequence data. These techniques included enzymatic manipulation, hybridization, molecular cloning, primer construction, and gel electrophoresis. After sequencing, I read and analyzed the raw sequence data in Sequencher and sent the novel sequence data to GenBank.

TEACHING EXPERIENCE**San Francisco State University****Fall 2020**

Instructor for Biology Colloquium

University of California Los Angeles**Fall 2020**

Co-Instructor for How Genetics has Contributed to Racial Injustice in America and World

University of California Los Angeles**Fall 2019**

Teaching Assistant for Population Genetics

University of California Los Angeles**Fall 2017**

Computational Instructor for Advanced Human Genetics

University of California Los Angeles**Fall 2016**

Teaching Assistant for Genetics Course

SKILLS

Programming Languages: Python, R, C++, MySQL

Operating Systems & General Applications: LINUX/UNIX, MS Windows, MS Office Suite

Computing Clusters & Software: AWS, Slurm, SGE

Bioinformatics Tools (selected): Next generation sequencing analyses (GATK, VCFTools, BCFTools, PLINK, SHAPEIT, Beagle), Phylogenetics (Sequencher, MEGA, FigTree, Geneious, Phylip), Data processing and visualization (tidyverse), Simulation software (ms, SLiM)

AWARDS AND HONORS

Stanford CEHG Fellow (\$13500 stipend & \$1500 travel fees)

2020-2021

SACNAS Scientific/STEM Symposia Session Workshop (Organized session)

2019

UCLA Dissertation Year Fellowship (\$21000 annual stipend & tuition and fees)

2019

Selected as a Platform Session moderator for the American Society of Human Genetics annual meeting

2018

Graduate Programs in Bioscience (GPB) Fellowship Incentive Program/Shine Fellowship (\$3000 annual)

2016-Present

National Science Foundation (NSF) Graduate Research Fellowship (GRFP) (\$34000 annual stipend & \$12000 fees)	2015-Present
Presidential Scholarship at University of New Mexico	2010-2014
New Mexico Lottery Success ABQ Scholarship	2010-2014
Undergrad Minority Scholarship	2013
Honors Award/Anthropology Scholarship	2013
Miscellaneous Scholarships	2012
Enterprise Rent-a-Car Diversity Scholarship	2011
Dean's List at University of New Mexico	2010-2014

CONFERENCES AND PRESENTATIONS

Mooney JA*, Huber CD, Service S, Sul JH, Marsden CD, Zhang Z, et al. Understanding the hidden complexity of Latin American population isolates.

- 2017 American Society of Human Genetics (ASHG) **Poster**
- 2018 American Society of Human Genetics (ASHG) **Poster** (Reviewer's Choice Abstract)
- 2018 Bay Area Population Genomics (BAPG) **Talk**
- 2018 Population Evolutionary and Quantitative Genomics (PEQG) **Poster**
- 2018 Informatics Center for Neurogenetics and Neurogenomics (ICNN) Symposium **Talk**
- 2018 UCLA Quantitative and Computational Biology Retreat **Talk**
- 2018 UCLA Human Genetics Retreat (Leena Peltonen Award Finalist) **Talk**

Mooney JA*, Yohannes A, Lohmueller KE. The impact of identity-by-descent on fitness and disease in domesticated and natural *Canid* populations. (*in review PNAS*)

- 2019 UCLA Human Genetics Retreat (Leena Peltonen Award Finalist) **Talk**
- 2020 Society for Molecular Biology and Evolution (SMBE) **Talk**

PUBLICATIONS

Mooney JA, Yohannes A, Lohmueller KE. The impact of identity-by-descent on fitness and disease in domesticated and natural *Canid* populations. [Preprint]. 2020 November 17. DOI:10.1101/2020.11.16.385443

Sura SA, Smith LL, Ambrose MR, Amorim CEG, Beichman AC, Gomez ACR, Juhn M, Kandlikar GS, Miller JS, **Mooney JA**, Mummah RO, Lohmueller KE, Lloyd-Smith JO. Ten simple rules for giving an effective academic job talk. PLoS Comput Biol. 2019 Jul;15(7):e1007163. PubMed PMID: 31344032; PubMed Central PMCID: PMC6657819.

Mooney JA, Huber CD, Service S, Sul JH, Marsden CD, Zhang Z, et al. Understanding the hidden complexity of Latin American population isolates. Am J Hum Genet. 2018;103(5):707–26. <https://doi.org/10.1016/j.ajhg.2018.09.013>.

Mooney JA, Marsden CD, Randall DA, Yohannes A, Wayne RK, Lohmueller KE. Inferring the demographic history and examining genetic diversity in a critically endangered canid: the Ethiopian wolf. (*in preparation*)

EXPERIENCE AND PROFESSIONAL ACTIVITIES

Student Advisory Committee Member for Graduate Programs in Bioscience (GPB) at UCLA	2018-2020
Organizing Committee, UCLA QCBio Annual Career Panel and Networking Event	2017
SACNAS Member at UCLA	2015-2017
Member of the Association for Multi-Ethnic Bioscientists' Advancement at UCLA	2015-2020
Founder and Chair of the Association for Multi-Ethnic Bioscientists' Advancement at UCLA	2015-2018
Primate Enrichment at Rio Grande Zoo in New Mexico	2010-2015

Jazlyn Mooney

Research Statement

For the last decade, large-scale genome-wide sequence data sets have provided valuable insights about complex traits in humans. The growth of genetics adds new opportunities to pose and answers questions about the genetic variation that contributes to disease phenotypes and a population's history, also known as demography. **I plan to pursue a career in academia, where the long-term goal of my research program is to develop methods to understand how demographic processes and evolutionary forces interact to affect genetic variation, in particular deleterious variation, in the human genome.** In addition to its biomedical relevance, my work will also increase representation of diverse populations in genomics research, which is imperative for ameliorating the diversity imbalances in current genomics data sets. Further, my research will bridge the fields of anthropology, biology, and statistics by using mathematical models to understand how populations evolve over time and to infer the evolutionary forces acting on modern genomes.

Proposed Research

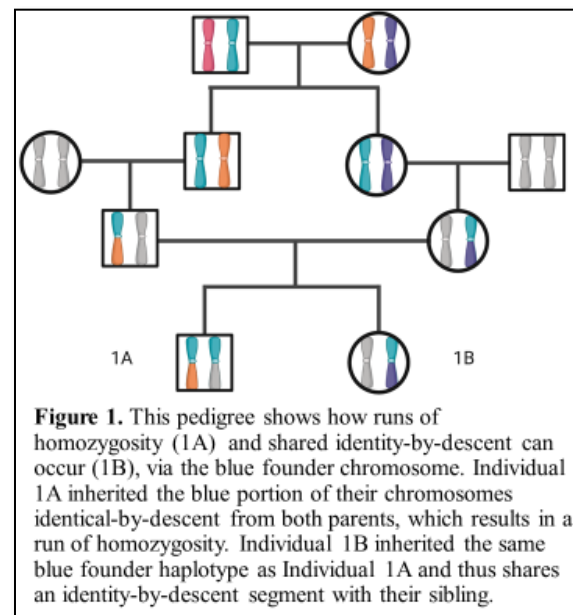
Since some deleterious variation may contribute to disease, it is important that scientists understand why and how deleterious variation integrates and remains within a population.

One of the core processes in population genetics is natural selection, by which deleterious genetic variants can be eliminated from populations. Given that natural selection is actively occurring, as a population geneticist, I am interested in the processes that have allowed for the persistence of deleterious variation. For example, we know that the demographic history of a population influences the distribution and prevalence of deleterious variation by affecting the efficiency of selection. Thus, deleterious variants can avoid being immediately eliminated from the genome by natural selection if selection is not strong or had enough time to effectively remove a variant.

My future research will use simulation frameworks, methods development, and empirical data analysis to do the following: 1) infer the demography of populations, 2) understanding the fitness impact of recessive variation, and 3) investigate complex trait architecture.

1a) Demographic History of Populations

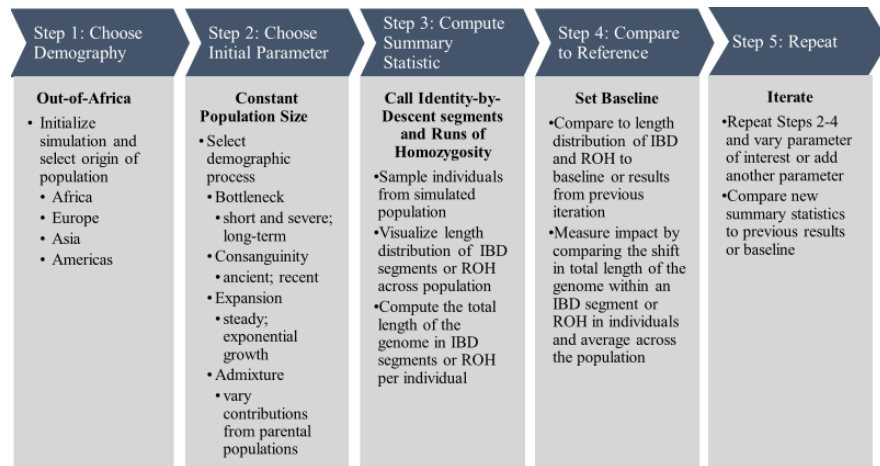
The demographic history of a population plays a key role in shaping the distribution of variation in the genome, however a population's history is not always known or documented. A large component of my future research will be dedicated to developing methods to determine the demographic history of a given population using a combination of simulations and empirical data. The simulation framework will be created to test the effect of multiple demographic processes on the length distribution of runs of homozygosity (ROH) and identical-by-descent segments (IBD). IBD segments can be viewed as segments of the genome that are inherited from a common ancestor and that are shared between at least two genomes in a population. ROH form when an individual inherits the same segment of their genome identically by descent from both parents (Figure 1). ROH can be viewed as a special case



of IBD, where IBD is simply within an individual rather than shared between individuals¹.

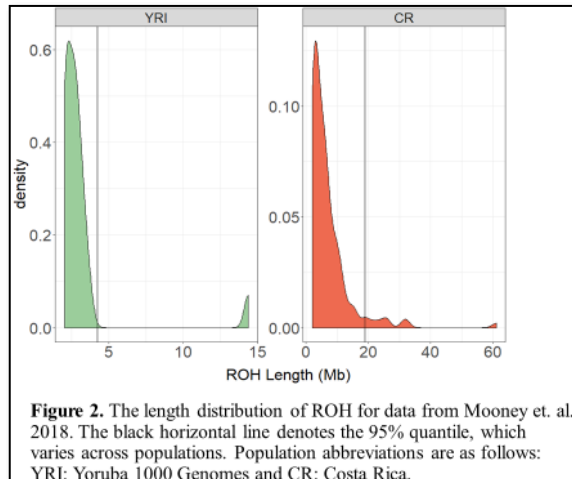
Using a combination of IBD and ROH makes it possible to capture both ancient and recent demography. Which will allow us to 1) obtain analytical predictions of how neutral population-genetic phenomena influence the length distribution of ROH and IBD, 2) better understand the length

distributions of ROH and IBD observed within worldwide human populations, 3) establish expectations for ROH and IBD given some knowledge of a population's demography, and 4) infer aspects of the population's demography given the empirical distribution of IBD and ROH. Additionally, this



approach is flexible so the models can also help to identify populations (human or non-human) with enrichments of IBD segments and ROH. Above, I have provided an example schematic that describes the processes for generating simulations and evaluating the impact of each demographic process. Models will be validated with empirical data where we have some knowledge of the population history (inferred or historical), and I will match the data back to the best-fitting demographic scenario that were simulated.

I hypothesize that demographic processes will differentially affect the length distribution of IBD and ROH. For example, some processes result in large amounts of ROH, while others will not. Thus, we



can observe differences in the length distributions of ROH. Evidence of these expectations can be seen in preliminary data from my previous work² (Figure 2). We can see that there is a clear difference in the length distribution of ROH across the two sampled populations. The YRI are expected to have very few long ROH and increased short ROH. This is because there likely has been no recent consanguinity, and the YRI did not experience the Out-of-Africa bottleneck. CR individuals carry larger amounts of long ROH than YRI, because of their demographic history, which includes a very recent founding event, subsequent bottleneck, and known

consanguinity. Given the availability of empirical data and historical information about populations, we can validate our simulations. This validation step will include populations from around the world with a wide range of demographic histories. I will use a mixture of publicly available data, such as the 1000 Genomes³, Human Genome Diversity Panel⁴, and data from my previous research, where a population's history is known and compare distributions of IBD and ROH to the simulated models.

After validation, we can apply the model to populations where demography was previously unknown. To this end, I will specifically focus on applying the results to populations that are underrepresented in current genetics data sets (i.e. non-European populations). The continued use of primarily European data sets results in poor predictions in non-European populations⁵. Continuing to broaden the diversity of populations in research is an essential component for a better understanding of biology, evolution, and for preventing further disparities in historically marginalized communities.

1b) Fitness Impact of Recessive Variation

After validating my models with demography only, I will extend these models to include selection. I will incorporate selection by requiring that nonsynonymous new mutations experience natural selection in accordance with the distribution of selection coefficients estimated in human populations⁶. I will examine the fitness consequences of deleterious variants in the context of ROH. Long ROH will be used exclusively because they are expected to not be strongly influenced by recombination and selection due to their recent origins^{2,7}. Further, due to the process by which ROH form, variants that may have been weakly deleterious in a heterozygous state, become homozygous and may now have a larger fitness impact. Thus, I hypothesize ROH will be enriched for deleterious variants with larger fitness effects, because consanguinity will increase the number of homozygous derived deleterious variants within long ROH.

I will test whether deleterious variants within long ROH have larger effects on fitness than variants outside of these regions in two ways: 1) computing fitness within long ROH versus outside long ROH, and 2) computing a mean selection coefficient within versus outside long ROH. I expect that long ROH will be enriched for variants with larger fitness effects because these ROH likely have not been subject to extensive selection or recombination, due to their recent origins. Given this knowledge, one could target those populations with increased IBD sharing and ROH for understanding disease architecture or use these regions to quantify the fitness impact of consanguinity. Results from this portion of my research program will be used to identify those populations that may be particularly fruitful for deciphering the fitness impacts of deleterious variants and their relationship with the architecture of complex human traits.

2) Complex Trait Architecture

Another avenue of my future research will be to better understanding complex trait architecture, where variants that contribute to the heritability of many traits remain unknown. There are multiple ways to identify variants that may be associated with a trait. One of the traditional models for association testing is SKAT⁸, where variants are weighted according to a beta distribution that depends on their minor allele frequency. However, this method was recently shown not to have enough power for rare alleles in realistic human demographies¹⁰. Thus, I propose that developing new demography informed models will allow us to detect more variant associations than traditional methods, like SKAT⁸, while maintaining a low false positive rate.

I will use models that incorporate demography and ROH, to increase the power of rare variant association tests. I will do this by weighting variants according to expectations provided by population genetic informed models. First, I will create a new statistic to compute the sum of variant effect sizes, within versus outside ROH in case-control data for complex traits. The idea behind this being that ROH could be particularly helpful to better understand disease architecture⁹ since ROH may harbor more recessive mutations that do not have full penetrance. These variants may not be significant genome-wide but have a large effect in aggregate, depending on the trait. Second, I will re-parameterize traditional SNP-effect sizes by coupling selection coefficients with trait effects. I will also examine how sample size affects the power to detect associations and what the lower bounds of sample size are for detecting associations with both the ROH based burden test and the new weighting scheme.

Previous and Current Research

During graduate school I worked on two relevant projects, my first project was focused on how demography affects the distribution of deleterious variation in a young, admixed population isolate². While most studies of population isolates have focused on populations where the founders all came from the same ancestral population, we investigated genomic diversity of recently admixed population isolates from Costa Rica and Colombia and compared their diversity to a benchmark population isolate, the Finnish. I found that the genetic diversity and genomic background of population isolates varies immensely. Specifically, admixture increased genetic diversity of the Colombian and Costa Rican

isolates. However, we still observed an enrichment of identity-by-descent (IBD) segments and a larger burden of runs of homozygosity (ROH) in the Latin American isolates relative to the Finnish. Our work suggests that there is no single genetic signature shared across all population isolates. Further, we determined that long ROH are generated by recent consanguinity, and that recent consanguinity has also led to an enrichment of derived deleterious variation in the Costa Rican and Colombian isolates. Thus, our results also provided a mechanism for how recent consanguinity can reduce fitness in natural populations. This work was published in *The American Journal of Human Genetics* (AJHG), the flagship journal of The American Society of Human Genetics.

I also used *Canids* (breed dogs, village dogs, and wolves) as a model system to test how recent demography and artificial selection affect the distribution of deleterious variation and the architecture of complex traits¹¹. These populations provide an excellent model for examining complex and recessive variants, because strong artificial selection during the breed formation period has simplified the genetic architecture of complex traits and allowed deleterious variation to persist in their genomes. We found that breed dogs show an enrichment of IBD and ROH relative to both village dogs and gray wolves. Further, we observed that in a sample of cases and controls for lymphoma, case-status is positively associated with the amount of the genome within a ROH; and that, on average, cases carried longer ROH than controls. We also used the distribution of ROH across the genome to identify regions of the genome with depletions of ROH as potential hotspots for inbreeding depression. We reported multiple loci where ROH are never observed. These loci are excellent candidates for assessing the fitness impacts of recessive disease-associated variants and for investigating the genetic architecture of regions of the genome where sharing two copies of the same variant can decrease fitness. We are revising this work after review at the *Proceedings of the National Academy of Sciences* (PNAS).

Currently, I am working on a project where I use inference based on genomic sharing to characterize admixture genealogically and to investigate an important quantity in understanding the nature of admixture processes, the number of individuals from each source population that have contributed to an admixed individual's genome. I use a combination of empirical data from African American populations¹² and extend mechanistic models of admixture¹³⁻¹⁶ to estimate how many distinct ancestors from each source population (Africans and Europeans/European Americans) on average are required to generate current admixture fractions within an individual. Approaching the admixture process in a demographically informed way allows for interpretation of these quantities in relation to African American history. This is particularly relevant since the African American genealogical tracing prior to the 1800's has been extremely limited due to forcible fracturing of enslaved families by enslavers, a practice of using only first names and not surnames for enslaved persons, lack of documentation of many of the enslaved even by first name in the written record, and a reticence of many formerly enslaved individuals to record genealogical information in the period after slavery¹⁷⁻¹⁹.

Taken together, my current and previous research has integrated data from both human and *Canid* populations, species with distinct demographic histories, to develop a more complete picture of mechanisms that shape patterns of genomic sharing and deleterious variation.

References

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4. Bergström, A., McCarthy, S.A., Hui, R., Almarri, M.A., Ayub, Q., Danecek, P., Chen, Y., Felkel, S., Hallast, P., and Kamm, J. (2020). Insights into human genetic variation and population history from 929 diverse genomes. *Science* 36, .
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6. Kim, B.Y., Huber, C.D., and Lohmueller, K.E. (2017). Inference of the distribution of selection coefficients for new nonsynonymous mutations using large samples. *Genetics* 206, 345–361.
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13. Verdu, P., and Rosenberg, N.A. (2011). A general mechanistic model for admixture histories of hybrid populations. *Genetics* 189, 1413–1426.
14. Goldberg, A., and Rosenberg, N.A. (2015). Beyond 2/3 and 1/3: the complex signatures of sex-biased admixture on the X chromosome. *Genetics* 201, 263–279.

15. Goldberg, A., Verdu, P., and Rosenberg, N.A. (2014). Autosomal admixture levels are informative about sex bias in admixed populations. *Genetics* 198, 1209–1229.
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Jazlyn Mooney

Teaching Philosophy

My commitment to teaching is strongly tied to my commitment to increase participation of historically marginalized students in STEM. Throughout graduate school and my postdoc, I have been both a teaching assistant and instructor for courses ranging from introductory genetics, population genetics, and a biology colloquium, to a course on how eugenics has influenced genetics. Through leading these courses, I have learned that in order for students to continue in STEM they must grasp both why we are interested in science and see themselves in a position where they can pursue science. A lot of the work for retention is therefore done in both the classroom and through mentorship.

For example, the graduate level course at SFSU is focused on introducing students to research from diverse faculty, postdocs, and graduate students across biology through a seminar series and student lunches. A portion of the seminar is dedicated to the speaker's career path, which is an excellent way to connect students with speakers. Students write a journal entry where they can discuss the science or what they thought was the most interesting part of the seminar. For most students, the discussion of career and life paths has been most interesting, especially since many students in my class are non-traditional (some work full time) and/or diverse (identify as belonging to an underrepresented ethnic group or as LGBTQ+). The ability to connect with the individual as a scientific role model agrees with much of the scientific teaching literature, which finds that professional identification plays a large role in the persistence and retention of diverse scientists in STEM.

As a multi-ethnic individual in science, I look forward to continuing my career in academia both through teaching courses in genomics, statistics, and biology as well as through mentoring. I will assess how my students and mentees are doing by conducting periodic surveys and I will use their feedback to improve. I will administer surveys to the students with the goal of ascertaining features of the course that best contributed to their plans for careers in STEM. Students will be surveyed at the beginning and end of the course about 1) what qualities describe a scientist, 2) whether they can identify a scientist who is similar to them, 3) what were three things they took away from the class. All these questions are relevant to science education, science teaching reform, and increasing retention of historically marginalized students in STEM.

My training in genetics, population genetics, anthropology, and computational biology has prepared me to design and teach courses related to computational genomics, evolutionary anthropology, and populational genomics. I could also assist in the general courses offered or specialized courses and seminars such Practical Analysis of Biological Data in R, Computational Genome Analysis, or Introduction to Genome Science. I would also be interested in continuing to teach a course about the relationship between eugenics and genetics. I was instructor for this course this past year and we formatted it to be accessible to undergraduate students and received very positive feedback. I believe it is important to inform future researchers about the origin of genomics, how its origins are linked to extremely racist views, and how genetics has been used to justify policies that maintain racial injustice. Once we understand the origins of genetics, we can discuss how scientists can and are working to make this field equitable. I think it also helps students understand why there is such trepidation from some individuals when it comes to participating in research, and how we can work to ameliorate these fears. This course would be an excellent fit in the pre-existing curriculum and could easily be shifted into a graduate level course.

I look forward to joining the USC community where I can continue honing my teaching and mentorship skills to make science feel more inclusive for historically marginalized students. Thank you for your time.

Jazlyn Mooney

Contributions to Diversity, Equity, and Inclusion

Growing up in Albuquerque in a college educated family, including a grandmother with a PhD, I assumed that most kids were like me. They grew up around books and science. Though we lived in a low-income area of town, I spent my elementary and middle school years in private schools. I noticed I was one of the only multi-ethnic (Black/Latina/Native American) kids at my school but most of my friends came from families with similar educational backgrounds. It was not until I entered public high school, that I realized how different my perception of the world was from reality for the average high-school student in my community. In 2010, the year I graduated, New Mexico had one of the worst (47th place) 4-year high school graduation rates in the country, with less than 70% of students graduated according to data from the National Center for Education Statistics. As I went on to my undergraduate degrees in Biology and Anthropology at the University of New Mexico, I did research and shadowed doctors. However, I also wanted to give back to my community, so I volunteered at our zoo and was a youth basketball and gymnastics coach. Afterwards, I went on to graduate school at UCLA where I studied genomics, with a focus on underrepresented (non-European) populations. I cultivated relationships with fellow historically marginalized students in STEM as both a peer and mentor, built student organizations, and worked within our Biosciences program increase recruitment and retention of our diversifying student body. I recently started my postdoc at Stanford, where I will continue studying underrepresented populations in genomics, working within postdoctoral diversity committees, and volunteering as a mentor.

In addition to teaching and mentoring students from historically marginalized groups, I also helped build an organization within UCLA to improve inclusivity for historically marginalized students. This organization is called the Association for Multi-Ethnic Bioscientists' Advancement (AMEBA). Our mission is to promote and increase retention of graduate students of historically underrepresented backgrounds in the biosciences. We achieve this mission by creating an inclusive community for graduate students that acknowledges and celebrates the diverse backgrounds of our members. Our group has grown from the original four founding members to approximately 40 active members, and 6 of our members have received the HHMI Gilliam Fellowship in the past three years. We perform outreach at local elementary schools and host quarterly meetings where we focus on coming together and celebrating the accomplishments of our members. We also organized and conducted a workshop at the Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS) in 2019, where we introduced participants to the basics of R-programming for data visualization and basic statistics. Three other members of AMEBA and I ran this workshop.

While at UCLA, I was also a member of the Student Advisory Committee to the Graduate Programs in Bioscience (GPB). GPB which consists of 14 departments and the committee advises home area directors and the director of GPB about current academic and financial climate for students. Our efforts resulted in the creation of emergency funding for students and relevant PI/departmental information for incoming students. We were working on implementing policies for parental leave upon my graduation. I also represented GPB at the Annual Biomedical Research Conference for Minority Students (ABRCMS) in 2017 and represented our Bioinformatics interdepartmental program at SACNAS in 2019.

As faculty, I look forward to continuing to take an active role in teaching and mentoring students. I am especially committed to making science more inclusive for those students and research participants who come from historically marginalized communities. I will do this by continuing my research on underrepresented populations in genomics, incorporating scientific-teaching practices (as detailed in my teaching statement), and creating a lab environment where my students feel welcome and can thrive.